```
> +
```

Funcii în Maple-sunt formate din numele funciei i argument/e

Funcia radical

```
> sqrt(x);
```

$$\sqrt{x} \quad (1.1.1)$$

```
> sqrt(49);
```

$$7 \quad (1.1.2)$$

Radical de ordin n

```
> surd(x,n);
```

$$\sqrt[n]{x} \quad (1.2.1)$$

```
> surd(125,3);
```

$$5 \quad (1.2.2)$$

```
> surd(8,3);
```

$$2 \quad (1.2.3)$$

```
> surd(-8,3);
```

$$-2 \quad (1.2.4)$$

```
> surd(1024,10);
```

$$2 \quad (1.2.5)$$

```
> surd(1024,10);
```

Funcia exponenial

```
> exp(x);
```

$$e^{x} \quad (1.3.1)$$

```
> exp(-3);
```

$$e^{-3} \quad (1.3.2)$$

Funcia exponenial cu baza a

```
> a^x;
```

$$a^{x} \quad (1.3.1.1)$$

```
> 10^2;
```

$$100 \quad (1.3.1.2)$$

```
> #Atenie!!! e^x nu are sens. Maple nu recunoate 'e' ca
  fiind Numrul lui Euler!
```

Funcia logaritmic

```
> log(x);
```

$$\ln(x) \quad (1.4.1)$$

```
> ln(x);
```

$$\ln(x) \tag{1.4.2}$$

```
> log[2](x);
```

$$\frac{\ln(x)}{\ln(2)} \tag{1.4.3}$$

```
> log[2](1024);
```

$$10 \tag{1.4.4}$$

Funcii trigonometrice

```
> sin(x);
```

$$\sin(x) \tag{1.5.1}$$

```
> sin(Pi/2);
```

$$1 \tag{1.5.2}$$

```
> #Atenie! Pi reprezint constanta 3.141592... , în timp ce
  'pi' scris cu liter mic este litera (simbolul)
> cos(x);
```

$$\cos(x) \tag{1.5.3}$$

```
> cos(2*Pi);
```

$$1 \tag{1.5.4}$$

```
> cos(3*Pi/2);
```

$$0 \tag{1.5.5}$$

```
> #Atenie!! Nu uitai operatorul * când avei o înmulire!
> tan(x);
```

$$\tan(x) \tag{1.5.6}$$

```
> cot(x);
```

$$\cot(x) \tag{1.5.7}$$

```
> cot(Pi/4);
```

$$1 \tag{1.5.8}$$

```
> arcsin(1);
```

$$\frac{1}{2}\pi \tag{1.5.9}$$

```
> arccos(1);
```

$$0 \tag{1.5.10}$$

```
> arctan(1);
```

$$\frac{1}{4}\pi \tag{1.5.11}$$

```
> arccot(1);
```

$$\frac{1}{4}\pi \tag{1.5.12}$$

Funcia modul

```
> abs(x);
```

$$|x| \tag{1.6.1}$$

Definirea unei funcii

```
> f:=x->a*x+b;
```

$$f := x \to a\,x + b \qquad (2.1)$$

```
> #returneaz, în notaie matematic, f(x)=a*x+b
> f(3);
```

$$3\,a + b \qquad (2.2)$$

Atribuirea de valori pt variabile date

```
> a:=5;
```

$$a := 5 \qquad (2.1.1)$$

```
> b:=-1;
```

$$b := -1 \qquad (2.1.2)$$

```
> f(3);
```

$$14 \qquad (2.1.3)$$

```
> #Atenie! a=5 si b=-1 vor rmâne în memorie. Dac tastm
  enter dup linia  f:=x->a*x+b; scris mai sus, va returna
  funcia cu valorile atribuite.
```

Descompunerea unei expresii algebrice

```
> expand((x-1)*(x+1));
```

$$x^2 - 1 \qquad (3.1)$$

```
> (x-1)*(x+1);
```

$$(x - 1)\,(x + 1) \qquad (3.2)$$

```
> #nu va face implicit înmulirea variabilelor
```

Grafice

Pachetul de ploturi

```
> with(plots);
```

[*animate*, *animate3d*, *animatecurve*, *arrow*, *changecoords*, *complexplot*, *complexplot3d*, *conformal*, *conformal3d*, *contourplot*, *contourplot3d*, *coordplot*, *coordplot3d*, *densityplot*, *display*, *dualaxisplot*, *fieldplot*, *fieldplot3d*, *gradplot*, *gradplot3d*, *implicitplot*, *implicitplot3d*, *inequal*, *interactive*, *interactiveparams*, *intersectplot*, *listcontplot*, *listcontplot3d*, *listdensityplot*, *listplot*, *listplot3d*, *loglogplot*, *logplot*, *matrixplot*, *multiple*, *odeplot*, *pareto*, *plotcompare*, *pointplot*, *pointplot3d*, *polarplot*, *polygonplot*, *polygonplot3d*, *polyhedra_supported*, *polyhedraplot*, *rootlocus*, *semilogplot*, *setcolors*, *setoptions*, *setoptions3d*, *spacecurve*, *sparsematrixplot*, *surfdata*, *textplot*, *textplot3d*, *tubeplot*] **(4.1.1)**

```
> #incarca pachetul de grafice
> with(plots):
> #punand ':' la finalul liniei de comand, nu va mai afisa pe
  ecran toate functiile incarcate
> #Atentie! Pachetul se incarca dupa fiecare instructiune
```

```
'restart;'
```

Curbe parametrice

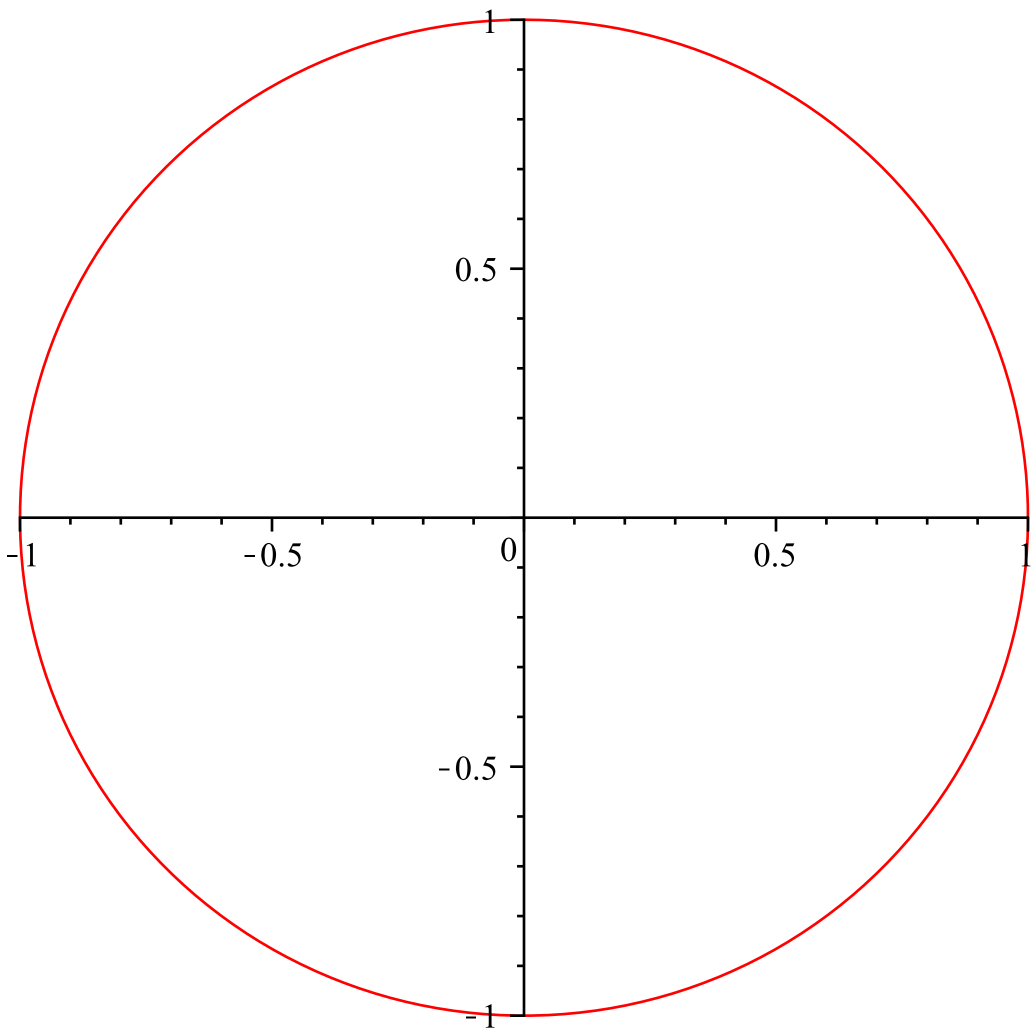
```
> restart;
> with(plots):
> f:=t->sin(t);
```

$$f := x \rightarrow \sin(x) \qquad \textbf{(4.2.1)}$$

```
> g:=t->cos(t);
```

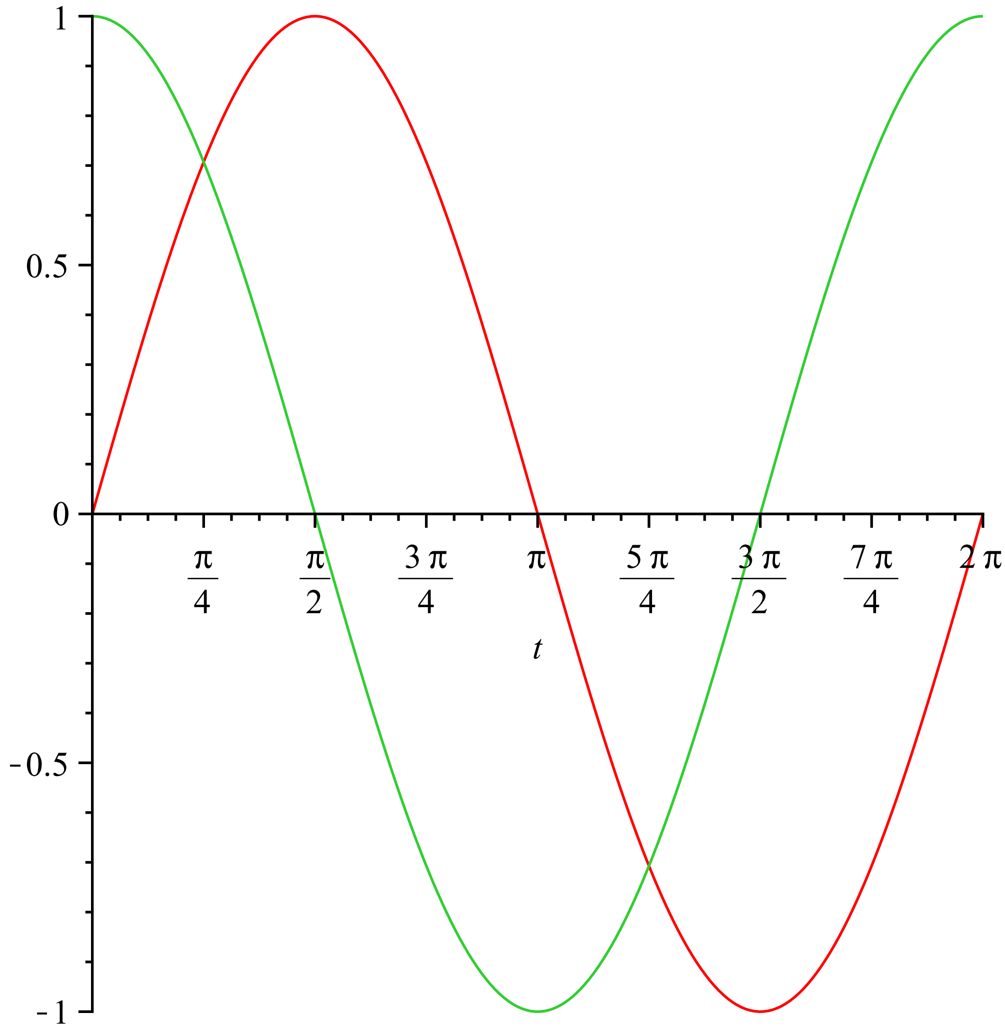
$$g := x \rightarrow \cos(x) \qquad \textbf{(4.2.2)}$$

```
> plot([f(t),g(t),t=0..2*Pi]);
```



Grafice de functii in aceeasi fereastra (Daca parametrul este setat in afara parantezei drepte, vor fi afiate 2 grafice independente

```
> plot([f(t),g(t)],t=0..2*Pi);
```




```
> #Argumentul t nu va mai lua valori simultan pt ambele functii
```

Siruri de curbe parametrice

```
> f:=(t,s)-> s*cos(4*t);
```

$$f := (t, s) \rightarrow s\cos(4\,t) \tag{4.3.1}$$

```
> x:=(t,s)->f(10*t,s)*cos(t);
```

$$x := (t, s) \rightarrow f(10\,t, s)\cos(t) \tag{4.3.2}$$

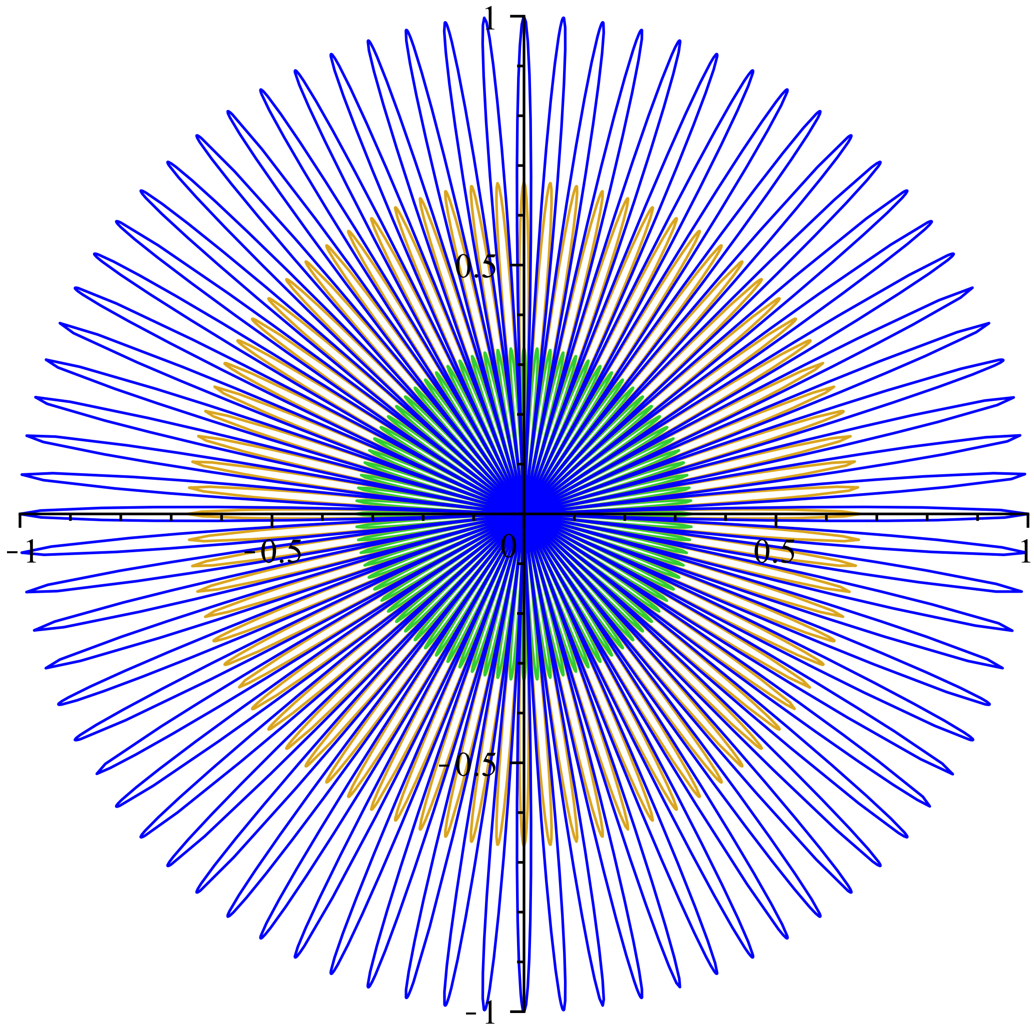
```
> y:=(t,s)->f(10*t,s)*sin(t);
```

$$y := (t, s) \rightarrow f(10\,t, s)\sin(t) \tag{4.3.3}$$

```
> list_xy:=[x(t,i/3),y(t,i/3), t=0..2*Pi]$i=0..3;
```

$$list\_xy := [0, 0, t = 0..2\,\pi], \left[\frac{1}{3}\cos(40\,t)\cos(t), \frac{1}{3}\cos(40\,t)\sin(t), t = 0..2\,\pi\right], \left[\frac{2}{3}\cos(40\,t)\cos(t), \frac{2}{3}\cos(40\,t)\sin(t), t = 0..2\,\pi\right], [\cos(40\,t)\cos(t), \cos(40\,t)\sin(t), t = 0..2\,\pi] \tag{4.3.4}$$

```
> plot([list_xy]);
```

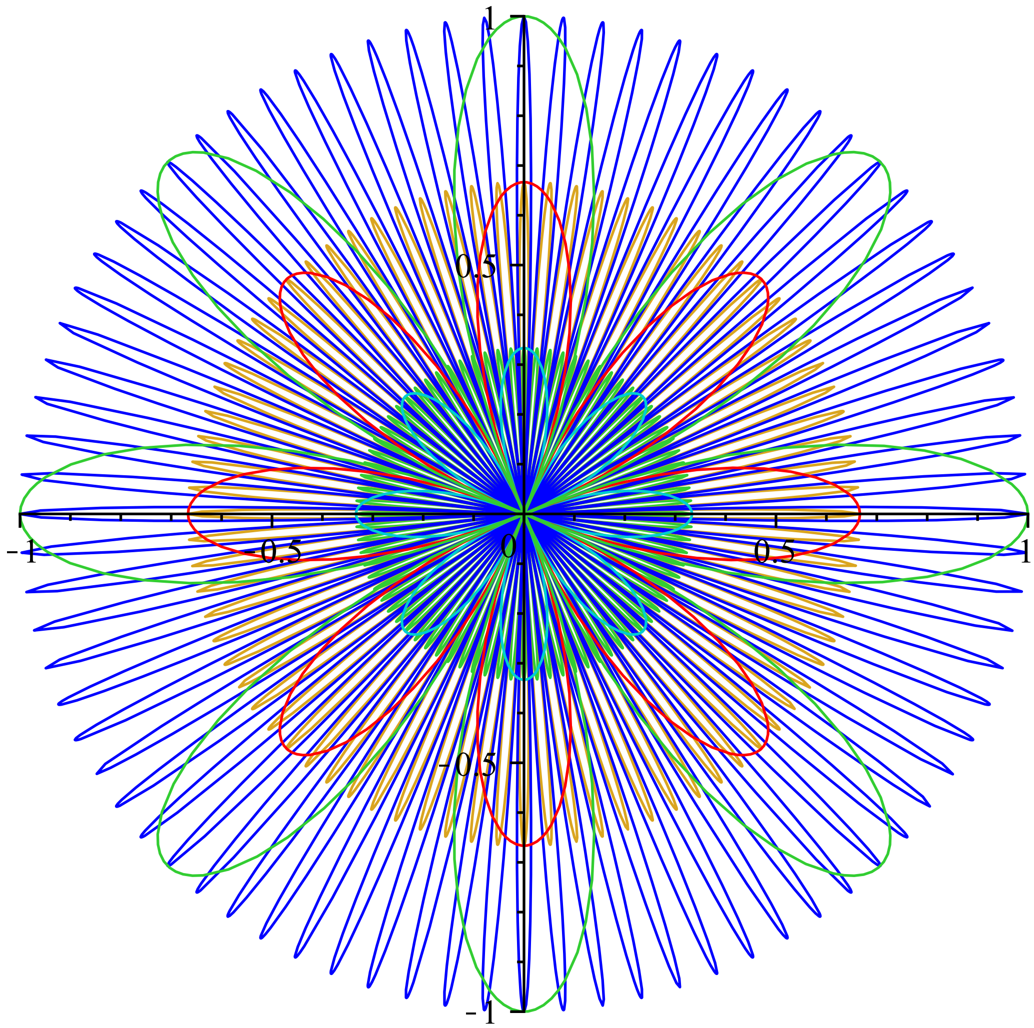



```
> u:=(t,s)->f(t,s)*cos(t);
```
$$u := (t, s) \rightarrow f(t, s)\cos(t) \quad \textbf{(4.3.5)}$$

```
> v:=(t,s)->f(t,s)*sin(t);
```
$$v := (t, s) \rightarrow f(t, s)\sin(t) \quad \textbf{(4.3.6)}$$

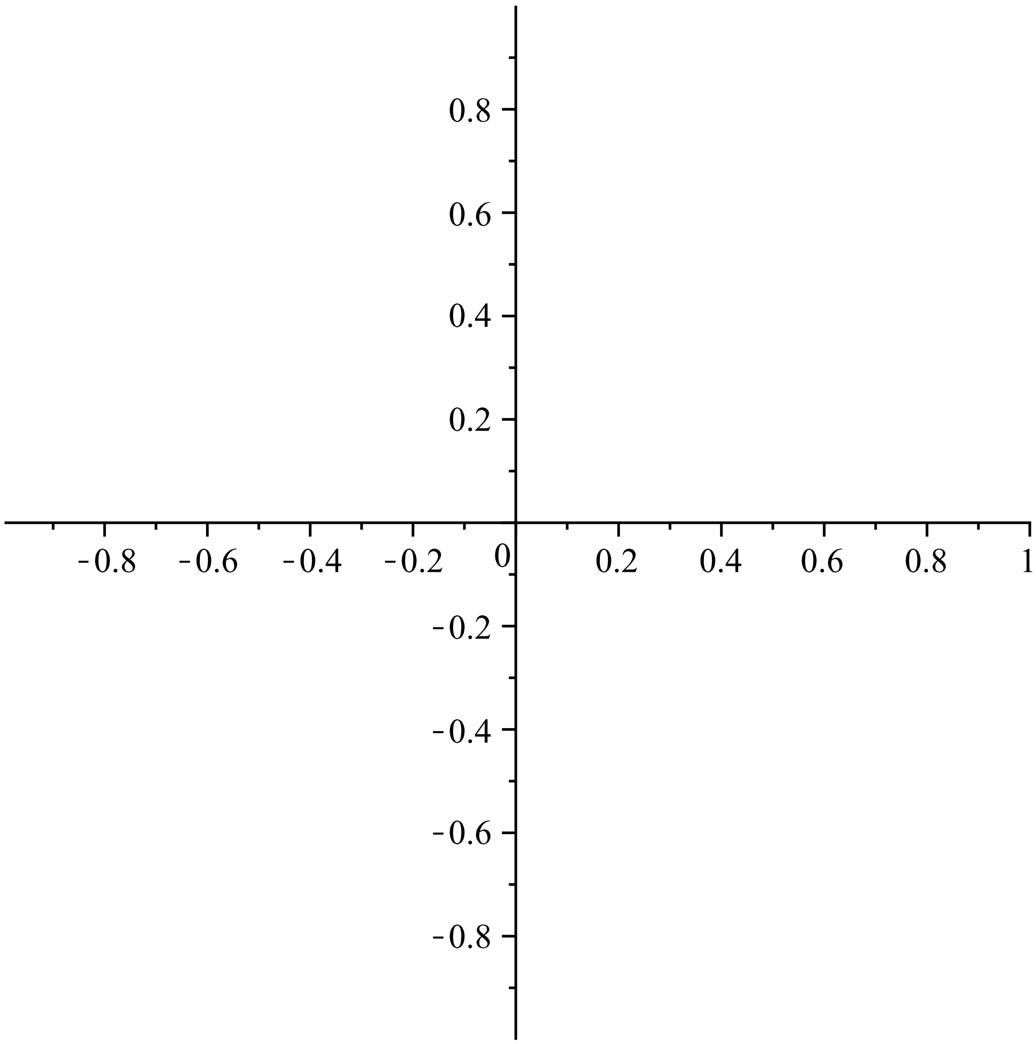
```
> list_uv:=[u(t,i/3),v(t,i/3), t=0..2*Pi]$i=0..3;
```
$$list\_uv := [0, 0, t = 0..2\pi], \left[\frac{1}{3}\cos(4t)\cos(t), \frac{1}{3}\cos(4t)\sin(t), t = 0..2\pi\right], \left[\frac{2}{3}\cos(4t)\cos(t), \frac{2}{3}\cos(4t)\sin(t), t = 0..2\pi\right], [\cos(4t)\cos(t), \cos(4t)\sin(t), t = 0..2\pi] \quad \textbf{(4.3.7)}$$

```
> plot([list_xy, list_uv]);
```



Dependenta de un parametru

```
> animate([x(t,s),y(t,s),t=0..2*Pi],s=-1..1,numpoints=1000,
  frames=50);
```




```
> #click dreapta pe grafic, Animation->Play
```